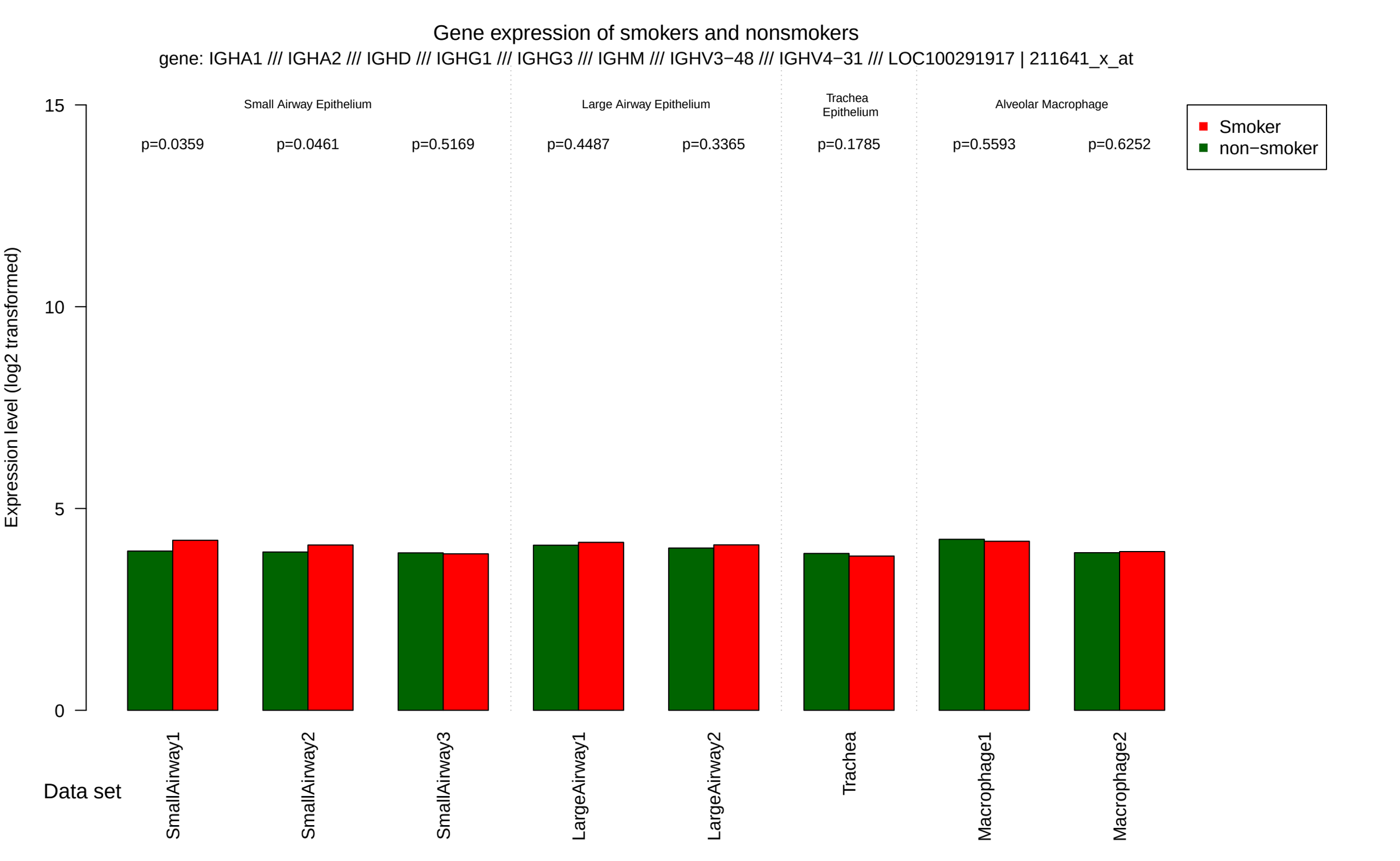
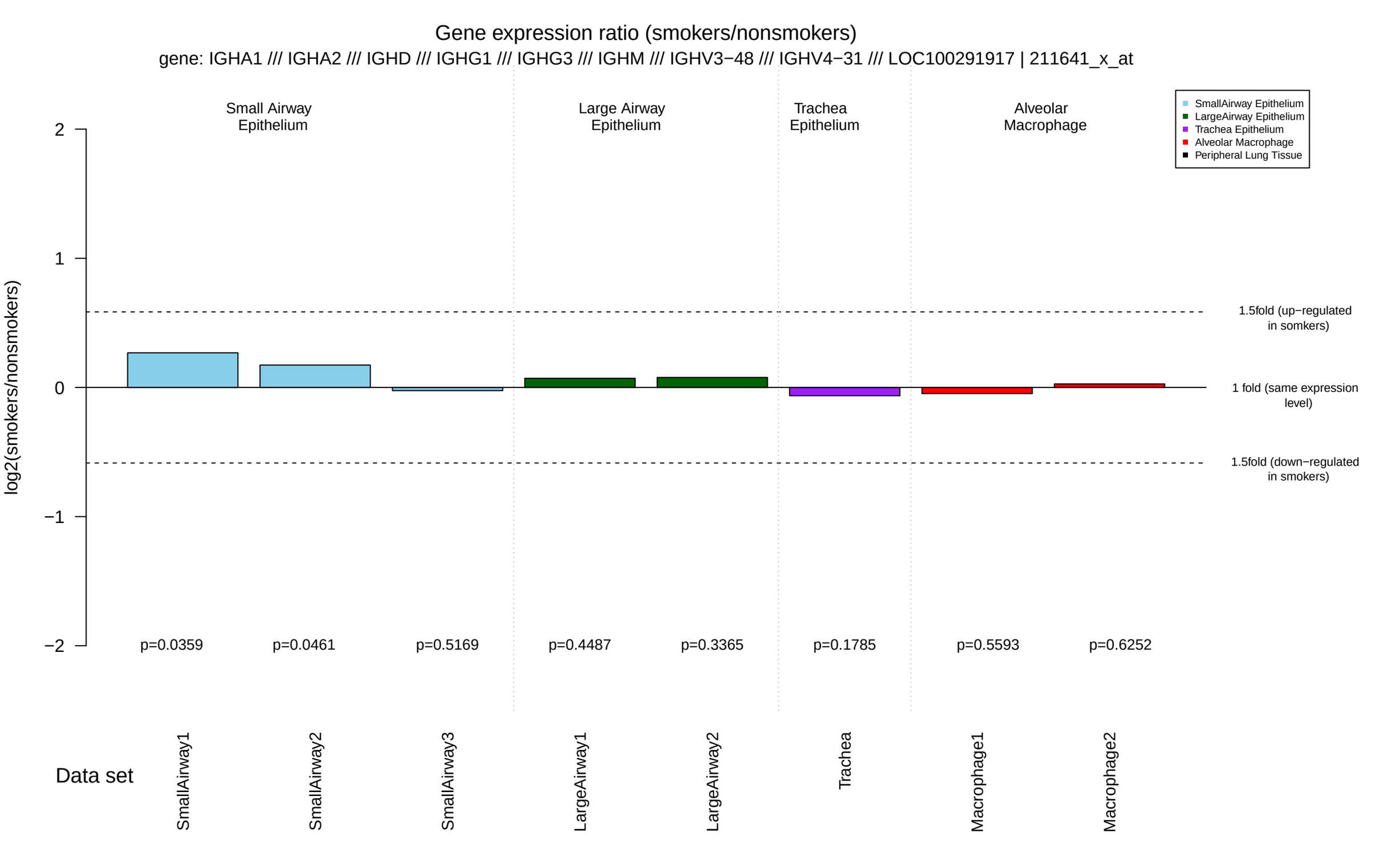
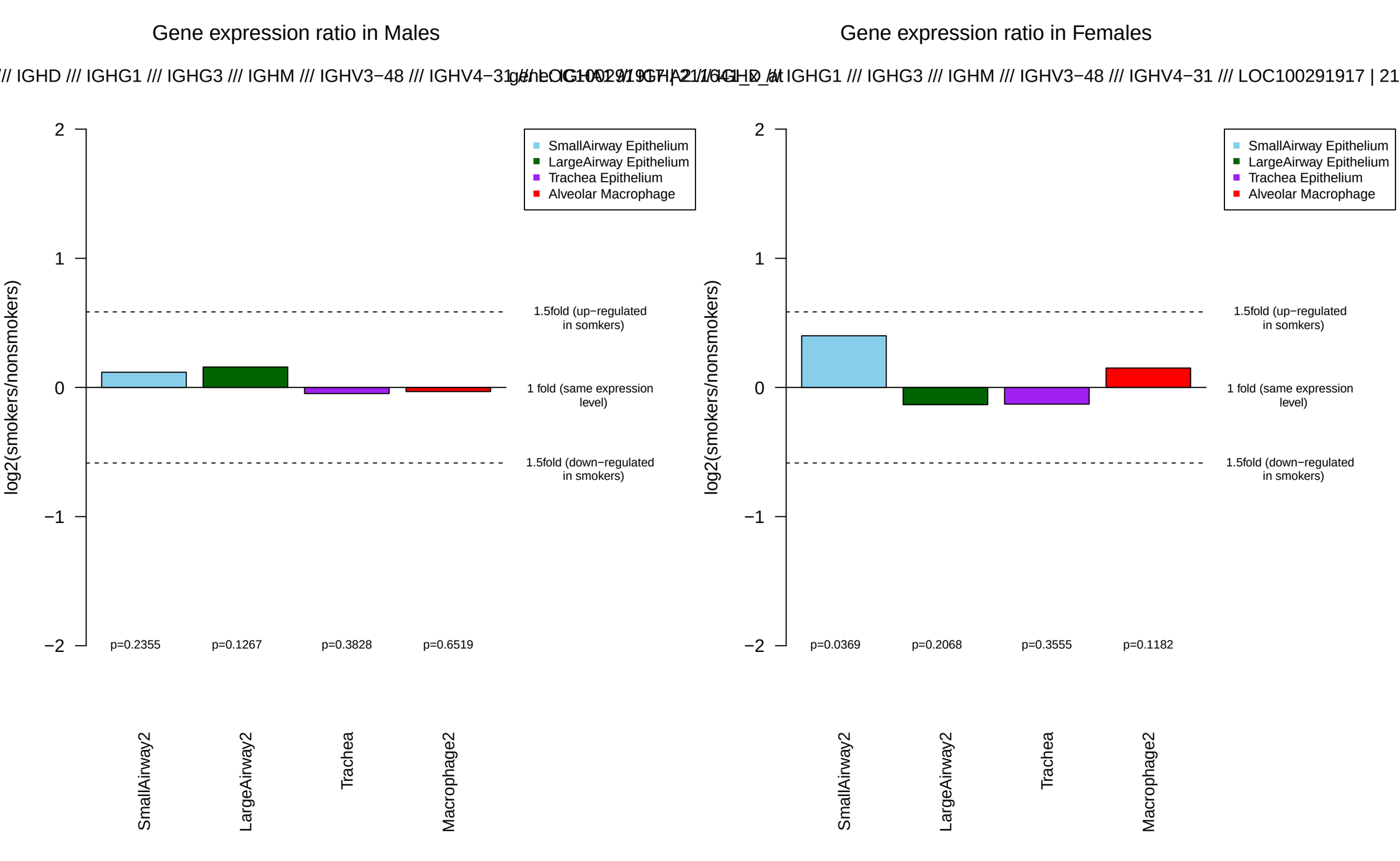
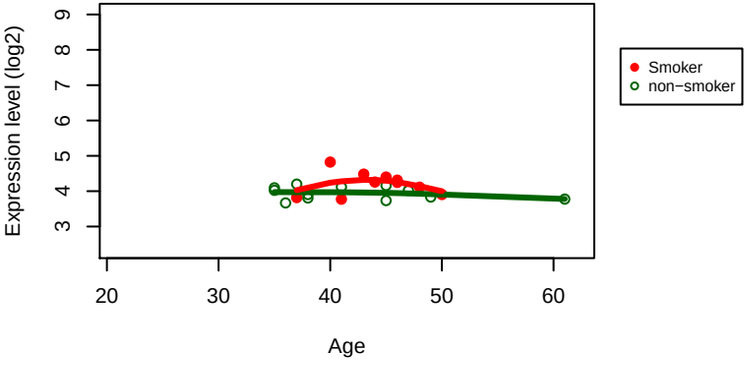

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC100291917 | 211641_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0359
p=0.0461
p=0.5169
p=0.4487
p=0.3365
p=0.1785
p=0.5593
p=0.6252
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

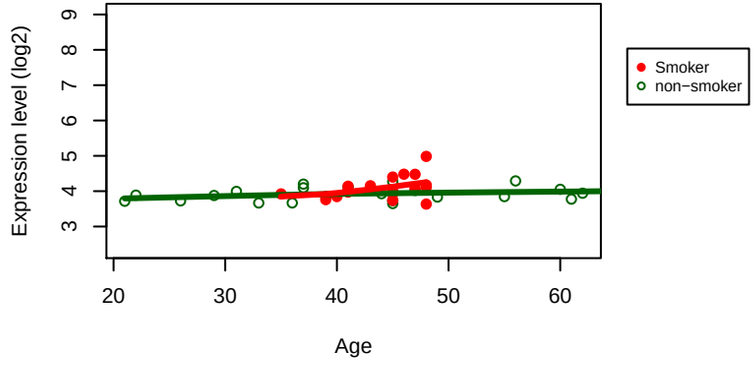

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC100291917 | 211641_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0359
p=0.0461
p=0.5169
p=0.4487
p=0.3365
p=0.1785
p=0.5593
p=0.6252
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio in Males
/// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC100291917 | 21164_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2355
p=0.1267
p=0.3828
p=0.6519
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
/// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC100291917 | 21164_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0369
p=0.2068
p=0.3555
p=0.1182
SmallAirway2
LargeAirway2
Trachea
Macrophage2

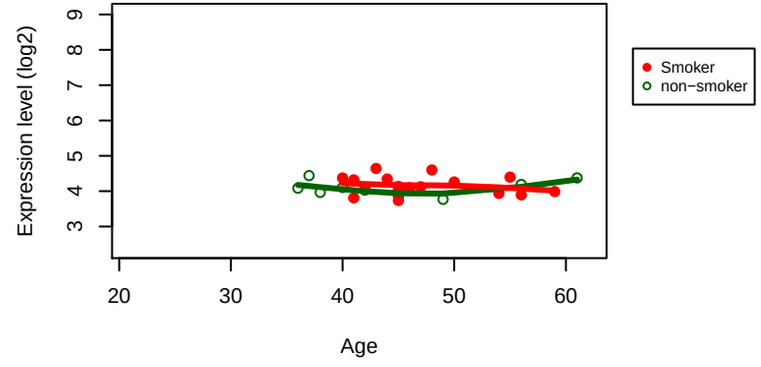
set: SmallAirway1 (p=0.0359)
IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC1002919



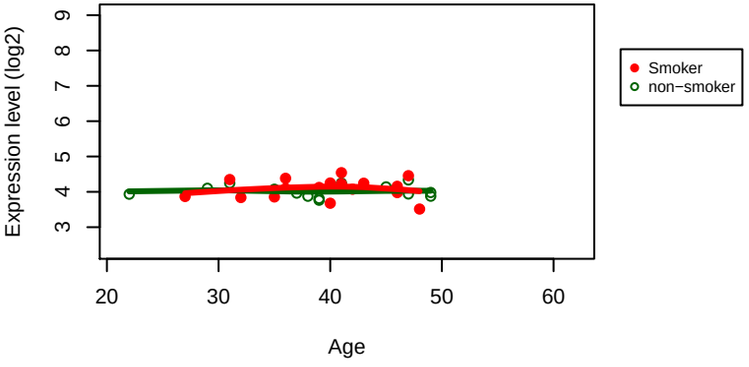
set: SmallAirway2 (p=0.0461)
IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC1002919



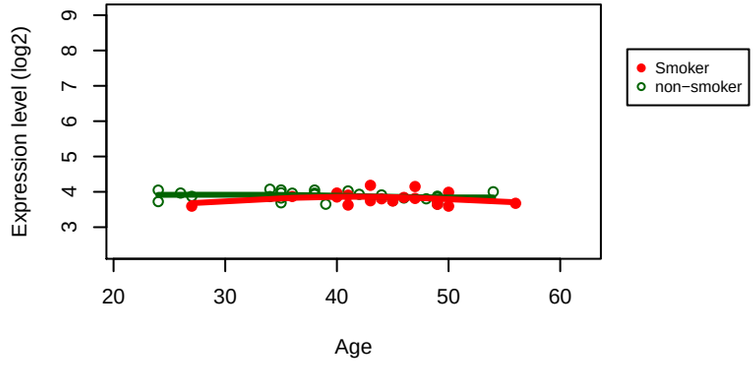
set: LargeAirway1 (p=0.4487)
IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC1002919



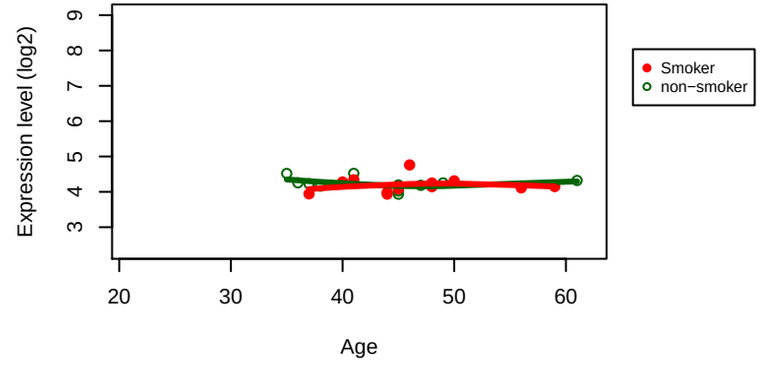
set: LargeAirway2 (p=0.3365)
IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC1002919



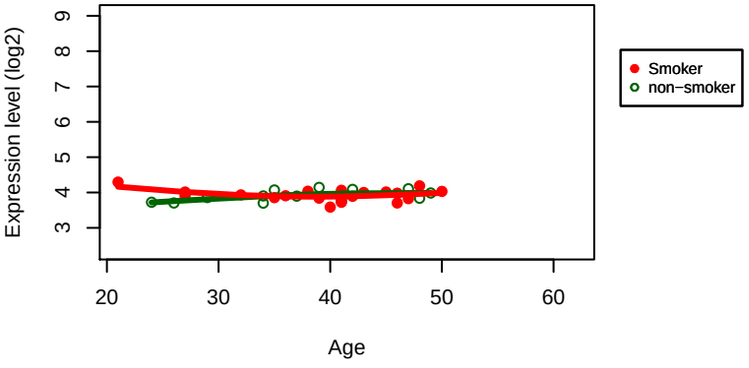
set: Trachea (p=0.1785)
IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC1002919

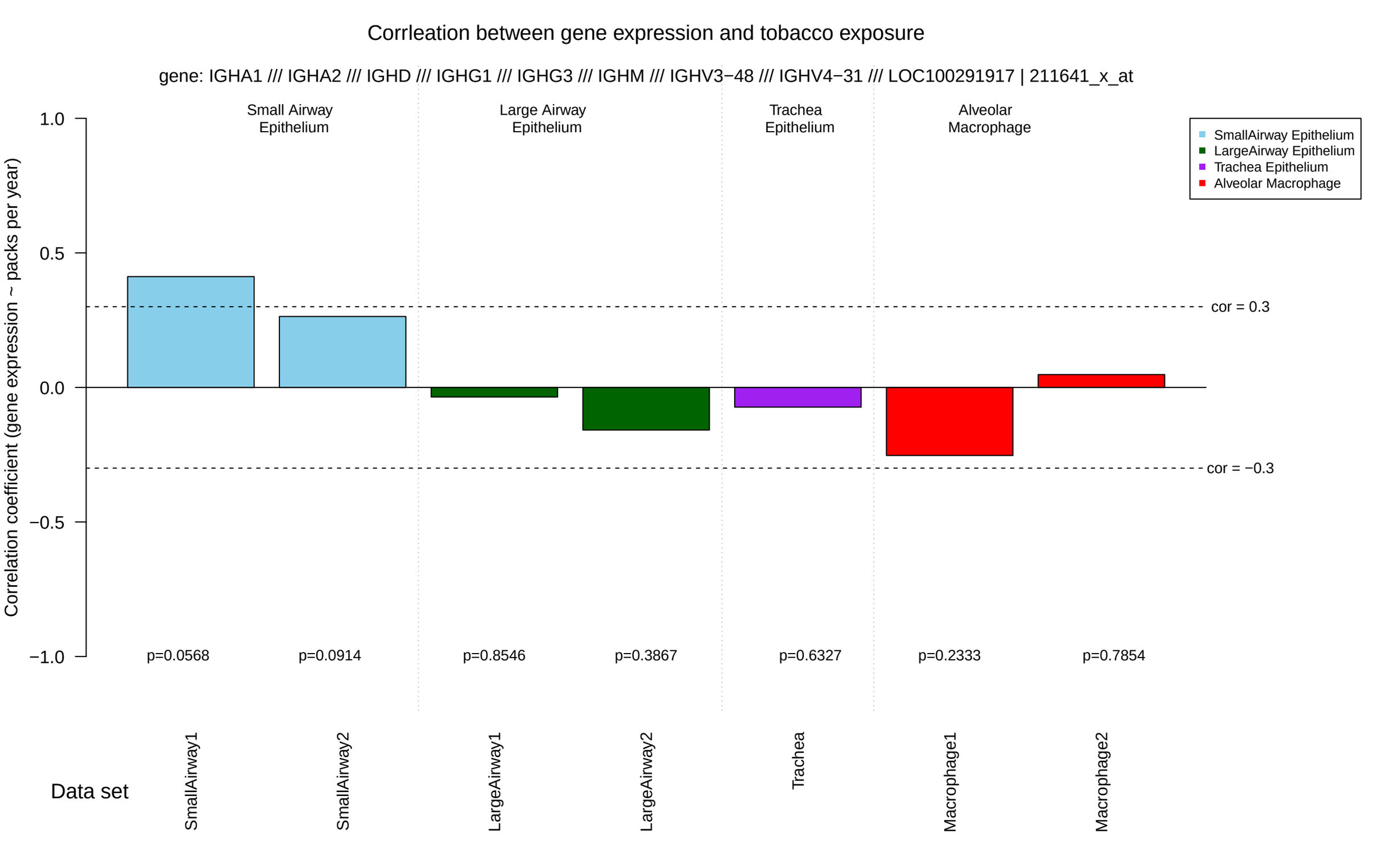


set: Macrophage1 (p=0.5593)
IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC1002919



set: Macrophage2 (p=0.6252)
IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC1002919




Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC100291917 | 211641_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0568
p=0.0914
p=0.8546
p=0.3867
p=0.6327
p=0.2333
p=0.7854
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2